Friction, Angled Forces and Tension

Answer key

Conventions used throughout

• **Forces:** F_g = gravity (F_g = mg, straight down), F_N = normal force, F_{app} = applied force, T = tension (as in the worksheet). F_s = static friction, largest value $F_s = \mu_s F_N$. F_k = kinetic friction, $F_k = \mu_k F_N$. F_f = friction on an object that is sliding (kinetic friction, $F_f = \mu F_N$).

• **Axes:** on a ramp, +x points down the ramp and +y points perpendicular to the ramp, away from it (as in 2.2). On flat ground, +x is the direction of motion and +y is up.

• **Signs in the working:** signed components, for example $F_{gy} = -mg\cos\theta$. The one exception is Example 2, which asks for a scalar expression: there every symbol is a size (always positive) and the minus signs are written into the equation.

• **Final answers:** accelerations as the scalar with its sign ($a = +7.0\ m/s^2$), then the vector with the axis it points along ($\vec{a} = 7.0\ m/s^2$ [positive x]). Other vectors get a direction in brackets, for example [forward].

• **Solve in letters first:** each unknown is isolated as a formula before the numbers go in.

• $g = 9.81\ m/s^2$. Every step keeps full precision; only final answers are rounded.

SIMULATION

Static and kinetic friction on an inclined plane

The following simulation is located at: https://ophysics.com/f2.html

1) What happens to the kinetic friction when you set the static friction to 0? Why might this be?

2) If we set kinetic friction to 0 but we have some static friction of some sort, what happens?

3) Notice that the static friction in this simulation will not exceed 1. There are cases though in real life where friction exceeds the coefficient of 1 though? Why do you think they restricted it?

4) Kinetic Friction is based on static friction. Notice how kinetic can never exceed static, why might this be?

1) What happens to the kinetic friction when you set the static friction to 0? Why might this be?

The kinetic friction drops to 0 too. The simulation never lets μ_k be bigger than μ_s (Question 4), so $\mu_s = 0$ forces $\mu_k = 0$. This matches real surfaces: it never takes more force to keep an object sliding than to start it sliding.

$$0 \leq \mu_k \leq \mu_s = 0 \quad \Rightarrow \quad \mu_k = 0$$

2) If we set kinetic friction to 0 but we have some static friction of some sort, what happens?

Static friction holds the box still while the ramp is shallow. Once the part of gravity along the ramp is bigger than the largest static friction, the box starts to slide. While it slides there is no friction at all, so it speeds up as if the ramp were frictionless:

Formula: $\Sigma F_x = ma_x$

Put in $\Sigma F_x = F_{gx} = mg\sin\theta$: $mg\sin\theta = ma_x$

Divide both sides by m: $a_x = g\sin\theta$

3) Why do you think they restricted it?

Most everyday pairs of surfaces (wood, metal, plastic, ice, snow) have coefficients less than 1. Values above 1 are special cases, such as soft rubber on dry pavement. A range of 0 to 1 covers the usual cases and keeps the simulation simple.

4) Notice how kinetic can never exceed static, why might this be?

At the moment an object starts to slide, the push equals the maximum static friction, $\mu_s F_N$. Once it slides, kinetic friction $\mu_k F_N$ takes over. Take + in the direction of the push:

Formula: $\Sigma F = ma$

Put in the forces: $F_{app} - F_k = ma$

Put in $F_{app} = \mu_s F_N$ and $F_k = \mu_k F_N$: $\mu_s F_N - \mu_k F_N = ma$

Divide both sides by m: $a = \dfrac{(\mu_s - \mu_k)F_N}{m}$

If kinetic friction were bigger ($\mu_k > \mu_s$), a would be negative: the net force would point backward, so the object would slow down and stop as soon as it started. Objects do keep sliding once started, so kinetic friction must be less than or equal to static friction: $\mu_k \leq \mu_s$.

FINAL ANSWER

1) Kinetic friction also goes to 0, because μ_k can't be bigger than μ_s

2) The box stays still until gravity along the ramp beats static friction, then it slides with no friction ($a = g \sin\theta$)

3) Most real surfaces have μ below 1, so 0 to 1 covers the usual cases

4) If μ_k were bigger, $F_{net} = (\mu_s - \mu_k)F_N$ would be negative and a sliding object would stop at once; objects keep sliding, so $\mu_k \leq \mu_s$

EXAMPLE 1

Static or kinetic?

Example 1) Identify the following as an example of static or kinetic friction (put an S for static a K for kinetic)

a. a sled moving along the snow

b. a fridge that you push on but it won't move

c. the friction between a car's tires and the road as it drives along

d. the friction between a car's tires and the road as skids to a stop

Ask one question each time: are the two surfaces sliding across each other? Sliding means kinetic friction (K). Not sliding means static friction (S).

Part	S or K	Why
a. sled	K	The sled slides over the snow.
b. fridge	S	The fridge doesn't move, so it isn't sliding on the floor.
c. tires, driving	S	A rolling tire doesn't slide: the part touching the road is at rest on it for an instant, then lifts off. That grip is what pushes the car forward.
d. tires, skidding	K	Locked tires slide along the road.

FINAL ANSWER

a. K

b. S

c. S

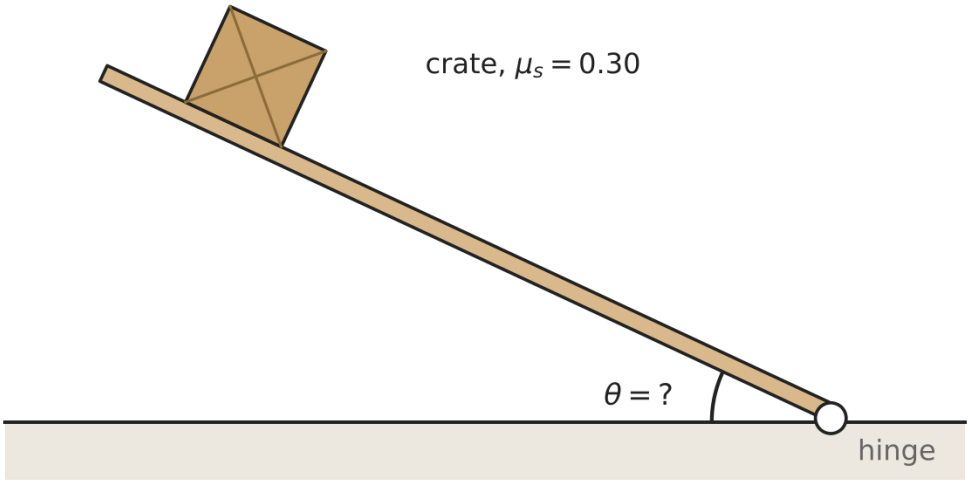
d. K

EXAMPLE 1

Overcoming static friction

Example 1) Overcoming Static Friction A crate is placed on an inclined board as shown. One end of the board is hinged so that the angle θ is adjustable. The coefficient of static friction between the crate and the board is 0.30. Determine the angle at which the crate just begins to slip.

Picture the situation

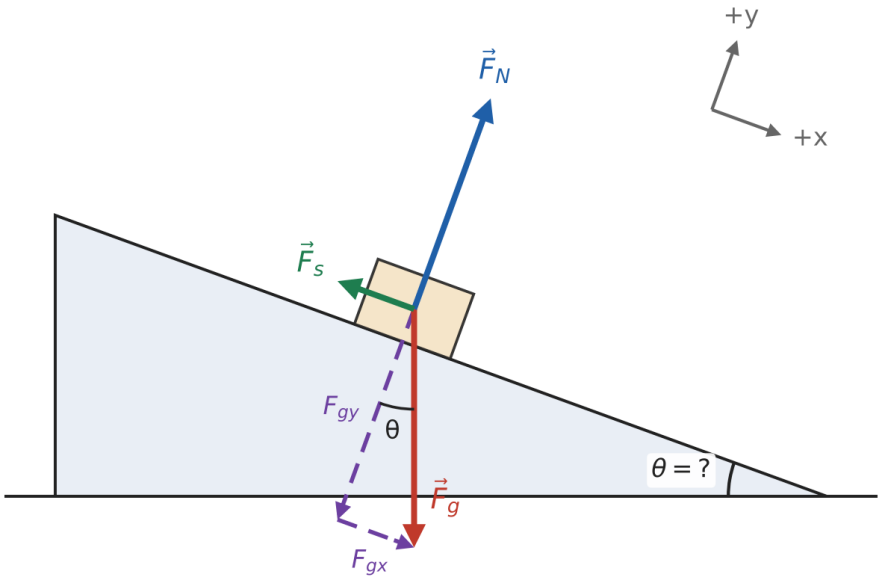


As the board is raised, gravity's pull down the board grows until static friction can't hold the crate any more.

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The angle θ at which the crate just begins to slip.

FBD of the crate just before it slips (+x down the board, +y away from the board)



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$\mu_s = 0.30$ $a_x = a_y = 0$ $\theta = ?$

The mass isn't given. Keep it as m; it will cancel.

Step 1 – Components of gravity

F_{gx} is opposite θ in the gravity triangle (sine) and points down the board (+x). F_{gy} is next to θ (cosine) and points into the board (−y):

Formula: $F_{gx} = mg\sin\theta$ $F_{gy} = -mg\cos\theta$

Step 2 – Normal force from the y direction

Formula: $\Sigma F_y = ma_y$

Put in the forces and $a_y = 0$: $F_N + F_{gy} = 0$

Put in $F_{gy} = -mg\cos\theta$: $F_N - mg\cos\theta = 0$

Add $mg\cos\theta$ to both sides: $F_N = mg\cos\theta$

Step 3 – Static friction is at its largest

"Just begins to slip" means static friction is as big as it can get, $F_s = \mu_s F_N$. It points up the board (−x):

Formula: $F_s = \mu_s F_N$

Put in $F_N = mg\cos\theta$: $F_s = \mu_s mg\cos\theta \Rightarrow F_{sx} = -\mu_s mg\cos\theta$

Step 4 – Balance the x direction and solve for θ

The crate is still at rest at this angle, so the x forces add to zero:

$$\text{Formula: } \Sigma F_x = ma_x$$

$$\text{Put in the forces and } a_x = 0: \quad F_{gx} + F_{sx} = 0$$

$$\text{Put in the components: } \quad mg\sin\theta - \mu_s mg\cos\theta = 0$$

$$\text{Add } \mu_s mg\cos\theta \text{ to both sides: } \quad mg\sin\theta = \mu_s mg\cos\theta$$

$$\text{Divide both sides by } mg: \quad \sin\theta = \mu_s \cos\theta$$

$$\text{Divide both sides by } \cos\theta: \quad \frac{\sin\theta}{\cos\theta} = \mu_s$$

From SOH CAH TOA, sin θ ÷ cos θ = (opposite/hypotenuse) ÷ (adjacent/hypotenuse) = opposite/adjacent = tan θ:

$$\tan\theta = \mu_s = 0.30$$

$$\theta = \tan^{-1}(0.30) = 16.70°$$

The mass cancelled, so any crate on this board starts to slip at the same angle.

FINAL ANSWER

θ = 17°

EXAMPLE 2

Two boxes joined by a string

Example 2) Two Body Problems Two boxes are attached by a light, inextensible (not stretchy) string and pulled along as shown below. The co-efficient of friction between the boxes and the surface is 0.25

a) State whether each of the following is true or false. Correct any that are false.

a. The x component of F is equal to the magnitude of T

b. The acceleration of both boxes are equal

c. the net force on both boxes are equal

d. All forces acting on box B are either vertical or horizontal

b) Draw a free body diagram of each box (draw in the space beside the diagram).

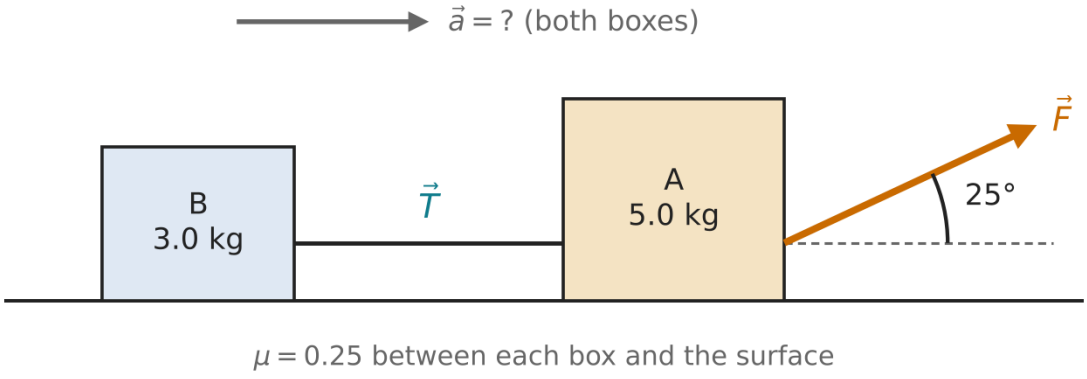
c) Write a scalar expression for the net force on the SYSTEM. (Ex: $F_{net} = F_{ap} - F_f$)

d) If the applied force is 75 N, determine the acceleration of the system (7.03 m/s^2)

e) Determine the magnitude of the tension in the string between the boxes. (28.5 N)

f) How much can the applied force increase before block A will start lifting off the ground?

Picture the situation



$\mu = 0.25$ between each box and the surface

The string is horizontal. F pulls box A at 25° above the horizontal, and the string drags box B along behind it.

Sign convention for this example: +x is the direction of motion (right) and +y is up. Every symbol is a size, and a minus sign in an equation means that force points the negative way.

(a) State whether each of the following is true or false. Correct any that are false.

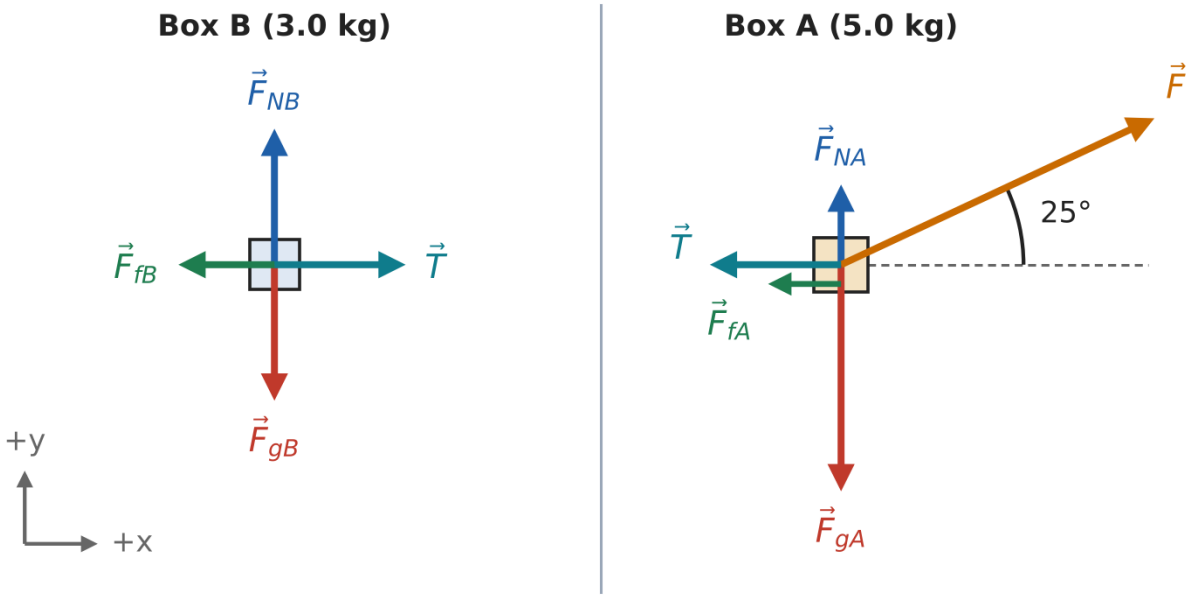
Statement	T or F	Why (and the correction)
a. The x component of F is equal to the magnitude of T	False	F_x has to speed up **both** boxes and beat the friction on both. T only has to do that for box B. **Correction:** the x component of F is greater than the magnitude of T.
b. The acceleration of both boxes are equal	True	The string doesn't stretch, so the boxes move together and speed up together.
c. the net force on both boxes are equal	False	Same a but different masses, and F_{net} = ma. **Correction:** the net force on A is greater than the net force on B (5.0 kg versus 3.0 kg).
d. All forces acting on box B are either vertical or horizontal	True	Gravity (down), normal force (up), tension (along the horizontal string) and friction (horizontal).

(b) Draw a free body diagram of each box (draw in the space beside the diagram).

Box A: gravity down, normal force up, F at 25° above the horizontal, tension back toward B, and friction backward (opposite the motion). F pulls partly upward, so it lifts some of A's weight off the floor and F_{NA} is smaller than F_{gA}.

Box B: gravity down, normal force up (same size, since nothing else is vertical), tension forward, and friction backward.

FBDs of each box (+x right, +y up)



(c) Write a scalar expression for the net force on the SYSTEM. (Ex: $F_{net} = F_{ap} - F_f$)

Step 1 – Which forces count for the system

Treat both boxes as one object. Tension pulls A backward and B forward by the same amount, so the two tension forces cancel inside the system. In the x direction the system feels the x part of F (forward) and friction on each box (backward):

$$F_{net} = F_x - F_{fA} - F_{fB}$$

Step 2 – Write each force in terms of what we know

F_x is next to the 25° angle (cosine). Each friction force is μF_N. For box A, the normal force is its weight minus the upward part of F, F sin θ (shown in (d)). For box B, the normal force is just its weight:

$$F_x = F\cos\theta \qquad F_{fA} = \mu(m_A g - F\sin\theta) \qquad F_{fB} = \mu m_B g$$

$$F_{net} = F\cos\theta - \mu(m_A g - F\sin\theta) - \mu m_B g$$

(d) If the applied force is 75 N, determine the acceleration of the system (7.03 m/s^2)

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$F = 75\ \text{N}$	$\theta = 25°$	$\mu = 0.25$
$m_A = 5.0\ \text{kg}$	$m_B = 3.0\ \text{kg}$	$a = ?$

Step 1 – Split F into components

F_x is next to the 25° angle (cosine) and F_y is opposite it (sine):

Formula: $F_x = F\cos\theta \qquad F_y = F\sin\theta$

$F_x = (75)\cos 25° = 67.97\ \text{N} \qquad F_y = (75)\sin 25° = 31.70\ \text{N}$

Step 2 – Normal force on each box

Neither box moves up or down, so the y forces on each box add to zero. For A, F_y helps the normal force hold the box up:

Formula: $F_g = mg$

$F_{gA} = (5.0)(9.81) = 49.05\ \text{N} \qquad F_{gB} = (3.0)(9.81) = 29.43\ \text{N}$

Box A, $\Sigma F_y = 0$: $\quad F_{NA} + F_y - F_{gA} = 0$

Subtract F_y and add F_{gA} on both sides: $\quad F_{NA} = F_{gA} - F_y$

$F_{NA} = 49.05 - 31.70 = 17.35\ \text{N}$

Box B, $\Sigma F_y = 0$: $\quad F_{NB} - F_{gB} = 0 \quad \Rightarrow \quad F_{NB} = 29.43\ \text{N}$

Step 3 – Friction on each box

Formula: $F_f = \mu F_N$

$F_{fA} = (0.25)(17.35) = 4.338\ \text{N} \qquad F_{fB} = (0.25)(29.43) = 7.358\ \text{N}$

Step 4 – Formula for the acceleration, then the numbers

Put the expression from (c) into Newton's second law for the whole system (+x forward) and isolate a before putting in any numbers:

Formula: $F_{net} = (m_A + m_B)\,a$

Put in (c): $F\cos\theta - \mu(m_A g - F\sin\theta) - \mu m_B g = (m_A + m_B)\,a$

Divide by $(m_A + m_B)$: $a = \dfrac{F\cos\theta - \mu(m_A g - F\sin\theta) - \mu m_B g}{m_A + m_B}$

$$a = \frac{67.973 - 4.338 - 7.358}{5.0 + 3.0} = \frac{56.28}{8.0} = +7.035 \text{ m/s}^2$$

The three numbers on top are F_x, F_{fA} and F_{fB} from Steps 1–3.

(e) Determine the magnitude of the tension in the string between the boxes. (28.5 N)

Look at box B by itself. Only two forces act on it sideways: tension forward and friction backward. Together they give B its acceleration:

Formula: $F_{net} = ma$

Put in box B's forces: $T - \mu m_B g = m_B a$

Add $\mu m_B g$ to both sides and factor: $T = m_B(a + \mu g)$

$T = (3.0)(7.035 + (0.25)(9.81)) = 28.46$ N

(f) How much can the applied force increase before block A will start lifting off the ground?

Step 1 – When does A lift off?

From (d), $F_{NA} = F_{gA} - F \sin\theta$. As F grows, its upward part grows and the floor pushes up on A less and less. A starts to lift off when the floor no longer pushes at all, $F_{NA} = 0$. (The tension is horizontal, so it doesn't change A's vertical forces.)

Put in $F_{NA} = 0$: $\quad 0 = F_{gA} - F\sin\theta$

Add $F\sin\theta$ to both sides: $\quad F\sin\theta = F_{gA}$

Divide both sides by $\sin\theta$: $\quad F = \dfrac{F_{gA}}{\sin\theta}$

$$F = \frac{49.05\ \text{N}}{\sin 25°} = 116.06\ \text{N}$$

Step 2 – The increase

$$\Delta F = 116.06\ \text{N} - 75\ \text{N} = 41.06\ \text{N}$$

FINAL ANSWER

(a) a. False (F_x is greater than T); b. True; c. False (the net force on A is greater); d. True

(b) FBDs above

(c) $F_{net} = F\cos\theta - \mu(m_A g - F\sin\theta) - \mu m_B g$

(d) $a = +7.0\ \text{m/s}^2$; $\vec{a} = 7.0\ \text{m/s}^2$ [positive x]

(e) T = 28 N

(f) It can increase by 41 N (to 1.2×10^2 N)

Worksheet answer: (d) 7.03 m/s^2; (e) 28.5 N
Why the answers differ: The physics is the same; only the rounding differs. The unrounded answers are $a = 7.035\ \text{m/s}^2$ and T = 28.46 N. Every given value (5.0 kg, 3.0 kg, 0.25, 75 N) has 2 significant figures, so the answers can only have 2: $7.0\ \text{m/s}^2$ and 28 N. The worksheet keeps a third figure that the data don't support. (Its 28.5 N does match the unrounded 28.46 N.)

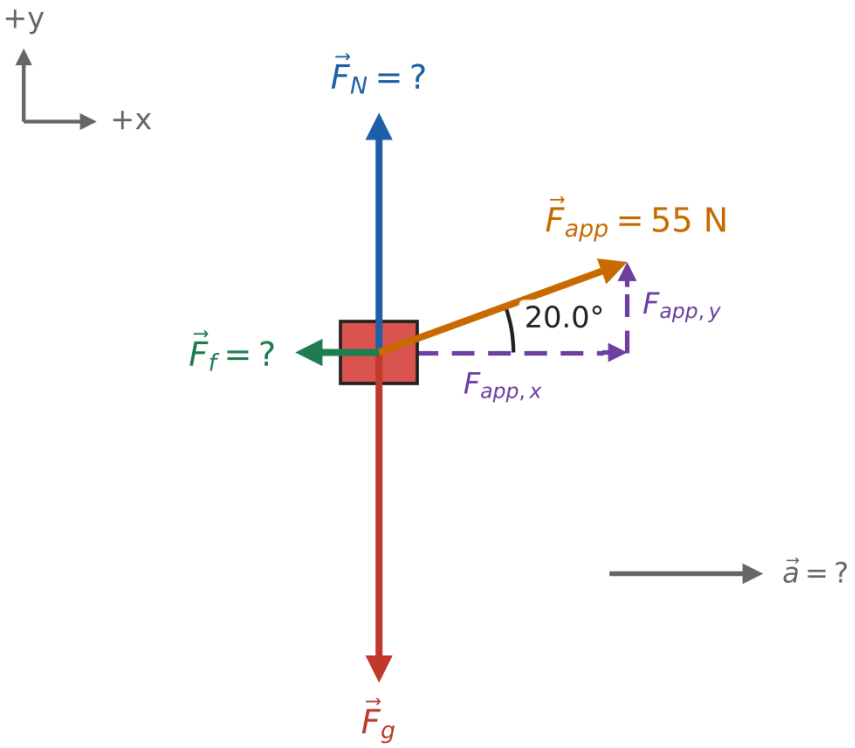
Check: box A gives the same acceleration: $F_x - T - F_{fA} = 67.97 - 28.46 - 4.338 = 35.17$ N, and $m_A a = (5.0)(7.0346) = 35.17$ N.

ADDITIONAL 1

Pulling a wagon at an angle

1. A child pulls a 7.0 kg wagon along a flat sidewalk with 55 N of force at an angle of 20.0° to the horizontal. If the co-efficient of friction is 0.35, determine the acceleration of the wagon. [4.9 m/s^2]

FBD of the wagon (+x forward, +y up)



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$m = 7.0$ kg	$F_{app} = 55$ N	$\theta = 20.0°$
$\mu = 0.35$	$a_y = 0$	$a_x = ?$

Step 1 – Components of the applied force

$F_{app,x}$ is next to the 20.0° angle (cosine). $F_{app,y}$ is opposite it (sine). Both are positive (forward and up):

$$F_{app,x} = (55)\cos 20.0° = 51.68 \text{ N}$$

$$F_{app,y} = (55)\sin 20.0° = 18.81 \text{ N}$$

Step 2 – Normal force

The wagon doesn't move up or down. The child's upward pull carries part of the weight, so the sidewalk pushes up less than mg:

Formula: $\Sigma F_y = ma_y$

Put in the forces and $a_y = 0$: $F_N + F_{app,y} + F_{gy} = 0$

$F_{gy} = -mg = -(7.0)(9.81) = -68.67$ N

Put in the numbers: $F_N + 18.81 - 68.67 = 0$

Add 68.67 and subtract 18.81 on both sides: $F_N = 49.86$ N

Step 3 – Friction

Friction points backward (−x):

Formula: $F_f = \mu F_N$

$F_f = (0.35)(49.86) = 17.45$ N $\Rightarrow$ $F_{fx} = -17.45$ N

Step 4 – Formula for the acceleration, then the numbers

Write each x force in letters, isolate a, then put in the numbers:

Formula: $\Sigma F_x = ma_x$

Put in the forces: $F_{app}\cos\theta - \mu(mg - F_{app}\sin\theta) = ma_x$

Divide both sides by m: $a_x = \dfrac{F_{app}\cos\theta - \mu(mg - F_{app}\sin\theta)}{m}$

$$a_x = \frac{(55)\cos 20.0° - (0.35)[(7.0)(9.81) - (55)\sin 20.0°]}{7.0}$$

$$a_x = \frac{51.68 - 17.45}{7.0} = \frac{34.23}{7.0} = +4.890 \text{ m/s}^2$$

FINAL ANSWER

a = +4.9 m/s^2

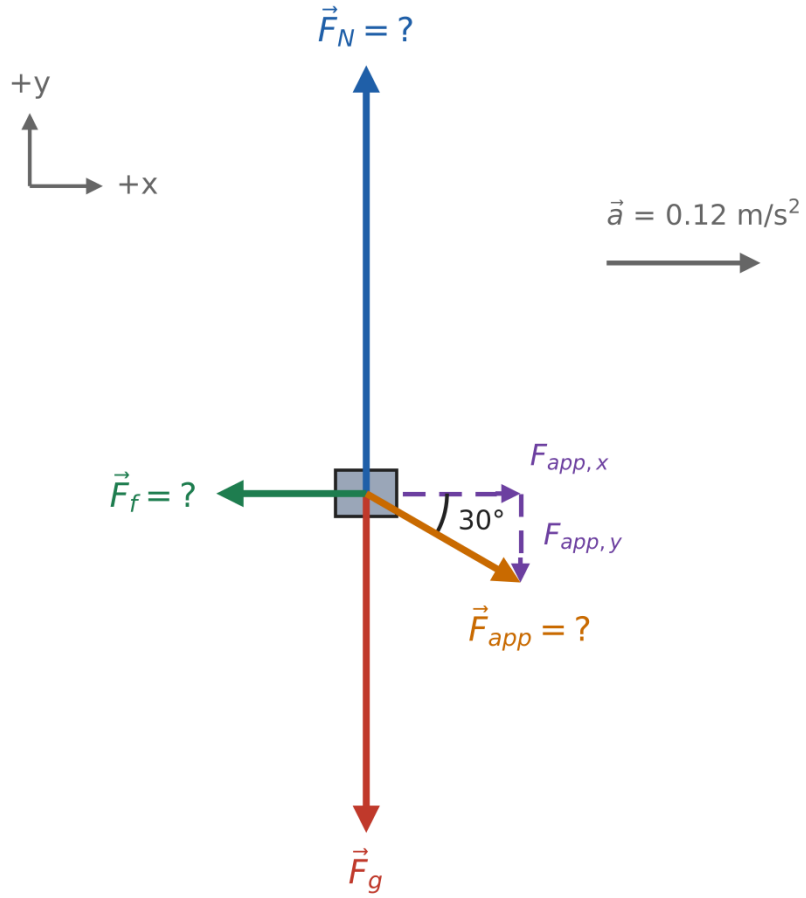
$\vec{a}$ = 4.9 m/s^2 [positive x]

ADDITIONAL 2

Pushing a shopping cart

2. A shopping cart is being pushed along the aisle of a grocery store with an acceleration of 0.12 m/s^2. The force is being applied along the handle of the cart at an angle of 30° below the horizontal. If the cart and contents have a mass of 12 kg and the co-efficient of friction is 0.35, determine the magnitude of the applied force. [62 N]

FBD of the cart (+x forward, +y up)



GIVEN

$m = 12$ kg	$a_x = 0.12$ m/s^2	$\theta = 30°$ (below)
$\mu = 0.35$	$a_y = 0$	$F_{app} = ?$

This time F_{app} is the unknown, and it shows up in both directions: its x part pushes the cart forward, and its y part pushes the cart into the floor, which makes the normal force and the friction bigger. Keep F_{app} as a letter until the end.

Step 1 – Components of the applied force

The handle points forward and down, so $F_{app,x}$ is positive and $F_{app,y}$ is negative:

$$F_{app,x} = F_{app}\cos 30° = 0.8660\,F_{app}$$

$$F_{app,y} = -F_{app}\sin 30° = -0.5000\,F_{app}$$

Step 2 – Normal force

Formula: $\Sigma F_y = ma_y$

Put in the forces and $a_y = 0$: $F_N + F_{app,y} + F_{gy} = 0$

$F_{gy} = -mg = -(12)(9.81) = -117.72$ N

Put in the numbers: $F_N - 0.5000\,F_{app} - 117.72 = 0$

Add $0.5000\,F_{app}$ and 117.72 to both sides:

$$F_N = 117.72 + 0.5000\,F_{app}$$

Step 3 – Friction

Friction points backward (−x):

Formula: $F_f = \mu F_N$

$F_f = (0.35)(117.72 + 0.5000\,F_{app}) = 41.202 + 0.1750\,F_{app}$

$F_{fx} = -(41.202 + 0.1750\,F_{app})$

Step 4 – Formula for F_{app}, then the numbers

First isolate F_{app} in letters:

Formula: $\Sigma F_x = ma_x$

Put in the forces: $F_{app}\cos\theta - \mu(mg + F_{app}\sin\theta) = ma_x$

Expand and move μmg across:

$$F_{app}\cos\theta - \mu F_{app}\sin\theta = ma_x + \mu mg$$

Factor out F_{app}: $F_{app}(\cos\theta - \mu\sin\theta) = m(a_x + \mu g)$

Divide: $$F_{app} = \frac{m(a_x + \mu g)}{\cos\theta - \mu\sin\theta}$$

$$F_{app} = \frac{(12)(0.12 + (0.35)(9.81))}{\cos 30° - (0.35)\sin 30°} = \frac{42.642}{0.6910} = 61.71 \text{ N}$$

FINAL ANSWER

F_{app} = 62 N

Check: $F_N = 117.72 + (0.5000)(61.71) = 148.57$ N, $F_f = (0.35)(148.57) = 52.00$ N, and $F_{app,x} - F_f = 53.44 - 52.00 = 1.44$ N = (12)(0.12).

ADDITIONAL 3

A locomotive pulling two cars

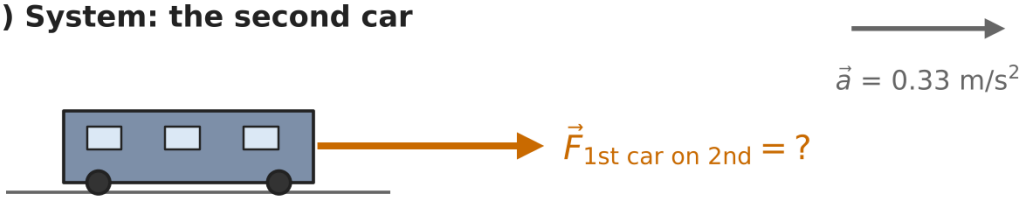
3. A train consisting of two cars pulled by a locomotive experiences an acceleration of 0.33 m/s^2 [fwd]. Friction is negligible. Each car has a mass of 3.1 x 10^4 kg.

   a. Determine the force exerted by the first car on the second car. (1.0 x 10^4 N)

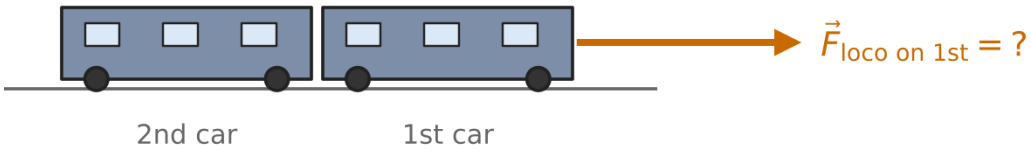
   b. Determine the force exerted by the locomotive on the first car. (2.0 x 10^4 N)

Choose the system so the force you want is the only sideways force on it (+x forward)

(a) System: the second car



(b) System: both cars



Gravity and the normal force cancel on every car (no vertical motion), and there is no friction, so only the pulls along the track matter. Every part of the train has the same acceleration.

(a) Determine the force exerted by the first car on the second car. (1.0 x 10^4 N)

The second car is pulled only by the first car, so that pull is its net force:

Formula: $F_{net} = ma$

$$F_{\text{1st on 2nd}} = (3.1 \times 10^4 \text{ kg})(0.33 \text{ m/s}^2) = 1.023 \times 10^4 \text{ N}$$

(b) Determine the force exerted by the locomotive on the first car. (2.0×10^4 N)

Take both cars as one system. The pull between the two cars is inside the system and cancels, so the locomotive's pull is the only sideways force, and it accelerates both cars:

Formula: $F_{net} = ma$

$$F_{\text{loco on 1st}} = (2 \times 3.1 \times 10^4 \text{ kg})(0.33 \text{ m/s}^2) = 2.046 \times 10^4 \text{ N}$$

The locomotive pulls twice as hard as the coupling between the cars, because it has to accelerate twice the mass.

FINAL ANSWER

(a) $\vec{F} = 1.0 \times 10^4$ N [fwd]

(b) $\vec{F} = 2.0 \times 10^4$ N [fwd]

ADDITIONAL 4

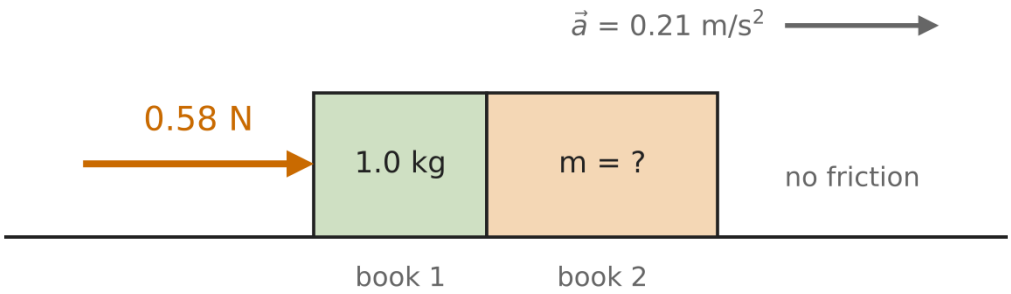
Two books pushed together

4. Two books are resting side by side, in contact, on a desk. An applied horizontal force of 0.58 N causes the books to move together with an acceleration of 0.21 m/s^2 horizontally. The mass of the book to which the force is applied directly is 1.0 kg. Neglecting friction, determine,

   a. The mass of the other book. (1.8 kg)

   b. The magnitude of the force exerted by one book on the other. (0.37 N)

Picture the situation



The 0.58 N push acts on book 1, and book 1 pushes book 2 ahead of it.

(a) The mass of the other book. (1.8 kg)

Step 1 – Formula for the second book's mass, then the numbers

Treat the two books as one system. The 0.58 N push is the only sideways force on it (the push between the books is inside the system):

Formula: $F_{net} = ma$

Put in $m = m_1 + m_2$: $F_{app} = (m_1 + m_2)\,a$

Divide by a and subtract m_1 : $m_2 = \dfrac{F_{app}}{a} - m_1$

$$m_2 = \frac{0.58}{0.21} - 1.0 = 2.7619 - 1.0 = 1.7619 \text{ kg}$$

(b) The magnitude of the force exerted by one book on the other. (0.37 N)

Book 2's only sideways force is the push from book 1, so that push is book 2's net force. By Newton's third law, book 2 pushes back on book 1 with the same size:

Formula: $F_{net} = ma$

$$F_{1 \text{ on } 2} = (1.7619 \text{ kg})(0.21 \text{ m/s}^2) = 0.3700 \text{ N}$$

FINAL ANSWER

(a) m = 1.8 kg

(b) F = 0.37 N

Check: book 1 alone: 0.58 N − 0.37 N = 0.21 N = (1.0 kg)(0.21 m/s^2).

ADDITIONAL 5

Figure skater pushing off the boards

5. A figure skater of mass m = 56 kg pushes horizontally for 0.75 s with a constant force against the boards at the side of a skating rink. Having started from rest, the skater reaches a maximum speed of 75 cm/s. Neglecting friction, determine the magnitude of,

a. The (constant) acceleration (1.0 m/s^2)

b. The force exerted by the skater on the boards (56 N)

c. The force exerted by the boards on the skater (56 N)

d. The displacement of the skater from the boards after 1.50 s (0.84 m)

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$m = 56$ kg

$v_1 = 0$

$v_2 = 75$ cm/s $= 0.75$ m/s

$\Delta t = 0.75$ s (pushing)

$a = ?$

$F = ?$

(a) The (constant) acceleration (1.0 m/s^2)

Acceleration is the change in velocity divided by the time it took:

Formula: $a = \dfrac{v_2 - v_1}{\Delta t}$

$$a = \frac{0.75 \text{ m/s} - 0}{0.75 \text{ s}} = 1.000 \text{ m/s}^2$$

(b) The force exerted by the skater on the boards (56 N)

The skater pushes on the boards, and by Newton's third law the boards push back on the skater. That push from the boards is the only sideways force on the skater, so it is the skater's net force. The two forces in the pair are the same size:

Formula: $F_{net} = ma$

$$F_{\text{boards on skater}} = (56\ \text{kg})(1.000\ \text{m/s}^2) = 56.0\ \text{N}$$

$$F_{\text{skater on boards}} = F_{\text{boards on skater}} = 56.0\ \text{N}$$

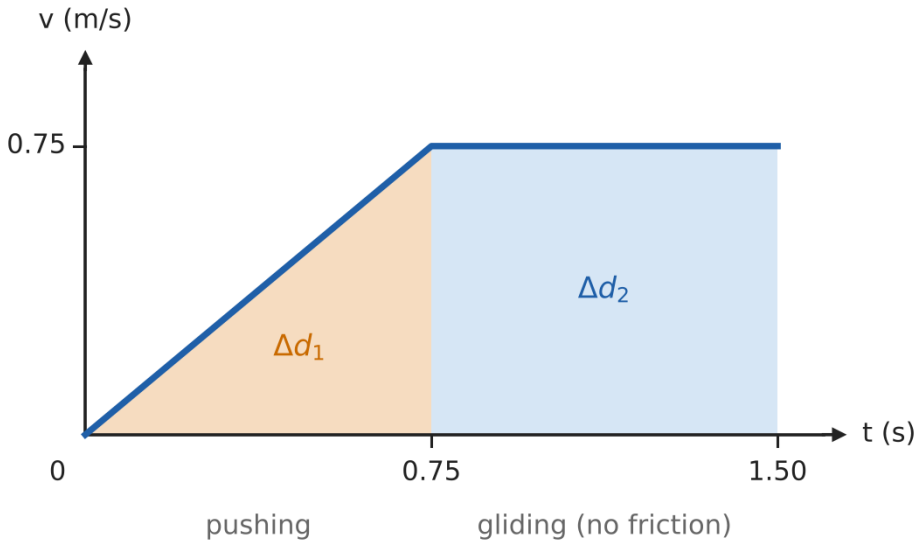
(c) The force exerted by the boards on the skater (56 N)

Found in (b): it is the force that accelerates the skater, 56.0 N.

(d) The displacement of the skater from the boards after 1.50 s (0.84 m)

The skater speeds up only while pushing (the first 0.75 s). After that there is no force (no friction), so by the first law the skater glides at a constant 0.75 m/s for the last 0.75 s. Find each distance and add them.

Velocity–time picture of the motion (the areas are the two distances)



Step 1 – Distance while pushing

Formula: $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

Put in $v_1 = 0$: $\Delta d_1 = \frac{1}{2} a \Delta t^2$

$\Delta d_1 = \frac{1}{2}(1.0 \text{ m/s}^2)(0.75 \text{ s})^2 = 0.2813 \text{ m}$

Step 2 – Distance while gliding

Constant velocity, so distance = velocity × time. The gliding time is 1.50 s − 0.75 s = 0.75 s:

Formula: $\Delta d = v \Delta t$

$\Delta d_2 = (0.75 \text{ m/s})(0.75 \text{ s}) = 0.5625 \text{ m}$

Step 3 – Add them

$\Delta d = \Delta d_1 + \Delta d_2 = 0.2813 + 0.5625 = 0.8438 \text{ m}$

FINAL ANSWER

(a) a = +1.0 m/s^2; $\vec{a}$ = 1.0 m/s^2 [positive x]

(b) 56 N

(c) 56 N

(d) Δd = 0.84 m

ADDITIONAL 6

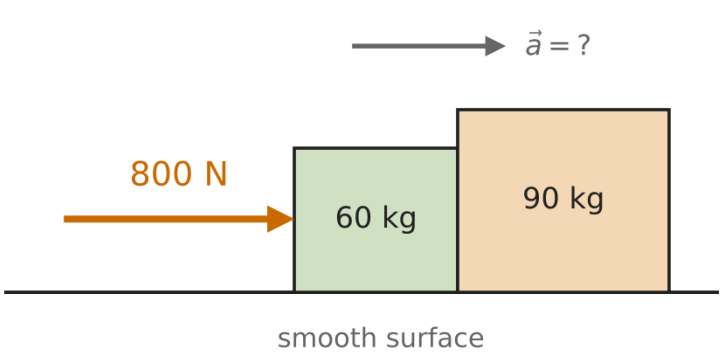
Pushing two boxes in contact

6. Two boxes, one with a mass of 60 kg and the other with a mass of 90 kg, are in contact and at rest on a smooth surface. An 800 N force is exerted on the 60 kg box toward the 90 kg box. Calculate:

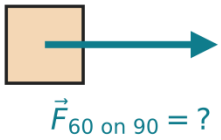
   a. the acceleration of the boxes. (5 m/s^2)

   b. the magnitude of the action and reaction forces between the boxes. (500 N)

The two boxes, and the sideways force on the 90 kg box (+x toward the 90 kg box)



90 kg box, sideways forces



"Smooth" means no friction. Each given value (60 kg, 90 kg, 800 N) has one significant figure, so the final answers are rounded to one.

(a) the acceleration of the boxes. (5 m/s^2)

Both boxes move together, so treat them as one system of 150 kg. The 800 N push is the only sideways force on that system:

Formula: $F_{net} = ma$

Divide both sides by m: $a = \dfrac{F_{net}}{m}$

$$a = \frac{800 \text{ N}}{60 \text{ kg} + 90 \text{ kg}} = \frac{800}{150} = 5.333 \text{ m/s}^2$$

(b) the magnitude of the action and reaction forces between the boxes. (500 N)

The 90 kg box is pushed only by the 60 kg box, so that push is its net force. By Newton's third law the 90 kg box pushes back on the 60 kg box with the same size:

Formula: $F_{net} = ma$

$F_{60\text{ on }90} = (90\text{ kg})(5.333\text{ m/s}^2) = 480.0\text{ N}$

FINAL ANSWER

(a) a = +5 m/s^2; $\vec{a}$ = 5 m/s^2 [positive x]

(b) 5 × 10^2 N on each box (the 60 kg box pushes the 90 kg box forward; the 90 kg box pushes back on the 60 kg box)

Check: 60 kg box alone: 800 N − 480.0 N = 320.0 N = (60 kg)(5.333 m/s^2).

ADDITIONAL 7

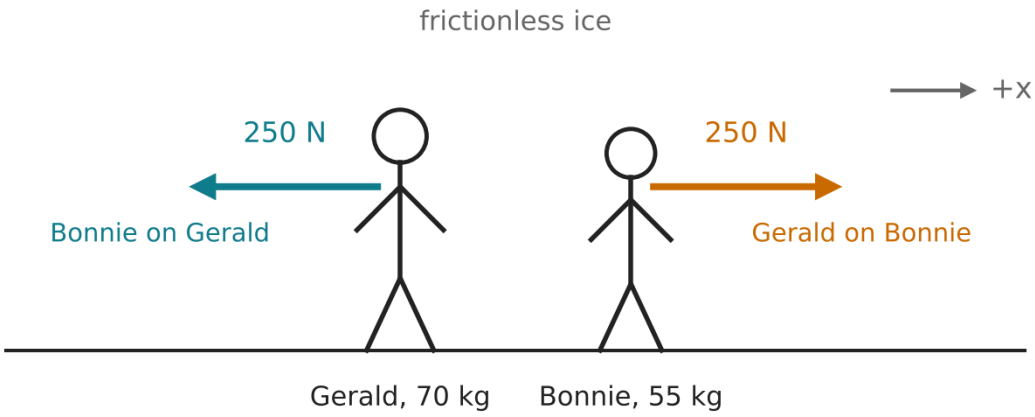
Bonnie pushes Gerald

7. A 55 kg girl, Bonnie, facing a 70 kg boy, Gerald, on a frictionless surface pushes Gerald with a force of 250 N for 0. 10 s. Calculate

a. Bonnie and Gerald's acceleration. (4.5 m/s^2 & -3.6 m/s^2)

b. Bonnie and Gerald's speed at the end of 0.50 s. (2.3 m/s & -1.8 m/s)

The third-law pair (+x is the way Bonnie moves)



(a) Bonnie and Gerald's acceleration. (4.5 m/s^2 & -3.6 m/s^2)

Step 1 – The forces

Bonnie pushes Gerald with 250 N. By Newton's third law, Gerald pushes Bonnie with 250 N in the opposite direction, so they both move, away from each other. Each person's only sideways force is this push:

$$F_{\text{G on B}} = +250 \text{ N} \qquad F_{\text{B on G}} = -250 \text{ N}$$

Step 2 – Each acceleration

Formula: $F_{net} = ma \quad \Rightarrow \quad a = \dfrac{F_{net}}{m}$

$$a_{\text{Bonnie}} = \frac{+250 \text{ N}}{55 \text{ kg}} = 4.545 \text{ m/s}^2$$

$$a_{\text{Gerald}} = \frac{-250 \text{ N}}{70 \text{ kg}} = -3.571 \text{ m/s}^2$$

Same size of force, but Bonnie has less mass, so she gets the bigger acceleration.

(b) Bonnie and Gerald's speed at the end of 0.50 s. (2.3 m/s & -1.8 m/s)

Step 1 – Velocity when the push ends (after 0.10 s)

They start at rest. The push, and so the acceleration, lasts only 0.10 s:

Formula: $v_2 = v_1 + a\Delta t$

$$v_{\text{Bonnie}} = 0 + (4.545)(0.10) = 0.4545 \text{ m/s}$$

$$v_{\text{Gerald}} = 0 + (-3.571)(0.10) = -0.3571 \text{ m/s}$$

Step 2 – From 0.10 s to 0.50 s

Once they stop touching, there is no force on either of them (the surface is frictionless). By the first law they keep the same velocity, so at 0.50 s their speeds are still the values from Step 1.

FINAL ANSWER

(a) Bonnie: $a = +4.5\ m/s^2$; $\vec{a}_{Bonnie} = 4.5\ m/s^2$ [positive x]

Gerald: $a = -3.6\ m/s^2$; $\vec{a}_{Gerald} = 3.6\ m/s^2$ [negative x]

(b) Bonnie: 0.45 m/s; Gerald: 0.36 m/s (moving apart)

Worksheet answer: (b) 2.3 m/s & −1.8 m/s
Why the answers differ: The worksheet's numbers are a × 0.50 s: (4.545)(0.50) = 2.3 m/s and (3.571)(0.50) = 1.8 m/s. That treats the 250 N push as lasting the whole 0.50 s. But the question says Bonnie pushes for only 0.10 s. Once their hands separate, no force acts on either of them (the surface is frictionless), so by Newton's first law they stop speeding up and keep the speeds they had at 0.10 s: 0.45 m/s and 0.36 m/s. Also, a speed is never negative: the minus sign on −1.8 only means Gerald moves the opposite way to Bonnie.